

Bike Sharing Services

Data Analysis Report

An end-to-end analysis of bike-sharing operations covering demand patterns, user behavior, station performance, and maintenance trends

50,000	5,000	200	136K
Total Trips	Total Users	Stations	Revenue

Tools Used: MySQL | Power BI
Dataset: 50,000 Trips · 7 Tables · 5 US Cities

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1. Project Overview

This project is a data analysis case study on a bike-sharing platform operating across five US cities — Boston, Chicago, New York, San Francisco, and Seattle. The main goal was to go through the data, clean it up, find patterns, and come up with useful insights that could help the business run better.

All the analysis was done using MySQL and the final results were presented in a Power BI dashboard. The work is divided into five stages — starting from checking data quality all the way to analyzing maintenance patterns.

Key Questions Explored

Area	Question Explored
Demand	When do most people ride? Which hours and days are busiest?
Stations	Which stations have too many or too few bikes?
User Behavior	How do casual riders and subscribers ride differently?
Maintenance	Which bikes break down most? Which stations are most affected?
Revenue	Where does the money come from — subscriptions or pay-per-ride?

2. About the Dataset

The dataset has seven tables. The central table is `trips_data` which records every bike ride taken. All other tables connect to it through shared IDs. Below is a quick summary of what each table contains:

Table	Records	What It Has
<code>trips_data</code>	50,000	Every trip — user, bike, start/end station, time, distance
<code>users_data</code>	5,000	User info — age, gender, subscription type, city
<code>stations_data</code>	200	Station details — name, city, capacity, coordinates
<code>bikes_data</code>	1,000	Bike info — type (standard or electric), last maintenance
<code>maintenance_data</code>	10,000	Repair logs — issue type, downtime hours, date
<code>revenue_data</code>	5,000	Payment info — fare amount, fines, payment method
<code>weather_data</code>	8,760	Hourly weather per city — temperature, rain, wind

`trips_data` is the largest and most important table. Most queries in this project start from `trips_data` and join to other tables to pull in extra details like station names, user subscription type, or maintenance history.

3. Stage 1 — Data Quality Check

Before starting the actual analysis I wanted to check whether the data was complete and reliable. This stage had three questions.

Q1 — How many records are in each table?

I counted the total records in all seven tables to check if any of them had unusually low numbers that might indicate missing or incomplete data.

Table	Total Records	Notes
trips_data	50,000	Main table — fully loaded
users_data	5,000	Matches expected user base
stations_data	200	All stations present
bikes_data	1,000	Full fleet captured
maintenance_data	10,000	Around 10 repairs per bike on average
revenue_data	5,000	One record per user
weather_data	8,760	365 days x 24 hours = full year of data

Key Finding: All tables are fully loaded with no missing records. `weather_data` has exactly 8,760 entries which is a complete year of hourly data.

Q2 — Are there missing values in `trips_data`?

I checked every important column in `trips_data` for null values since this is the main table and missing values here would affect almost everything else.

Column	Missing Values	% Missing	Impact
User_ID	0	0.00%	No issue
Bike_ID	0	0.00%	No issue
Start_Station_ID	0	0.00%	No issue
End_Station_ID	0	0.00%	No issue
Start_Time	0	0.00%	No issue
End_Time	0	0.00%	No issue
Trip_Duration_Min	0	0.00%	No issue
Distance_KM	0	0.00%	No issue

Key Finding: Zero missing values in trips_data. The dataset is clean and complete.

Q3 — Where are User IDs missing, and which ride type is affected?

I checked whether missing User IDs were mostly coming from casual rides (which could be guest checkouts) or from subscribers (which would indicate a bug).

Ride Type	Missing User IDs	Total Trips	Missing %
Casual	0	25,074	0.00%
Monthly	0	14,977	0.00%
Yearly	0	9,949	0.00%

Key Finding: No missing User IDs at all across any ride type. Casual riders make up the largest group with 25,074 trips — that is 50% of all rides.

4. Stage 2 — Data Cleaning

This stage was about finding and fixing any data problems before moving into the actual analysis. I checked for duplicates, missing IDs, bad trip durations, and inconsistent station names.

Q1 — Duplicate Trip Records

I looked for trips that had the same Trip ID, Start Time, User ID, and Bike ID appearing more than once — which would inflate trip counts and skew the analysis.

Key Finding: No duplicate trips found. All 50,000 records are unique.

Q2 — Handling Missing User IDs

The plan was to assign "CASUAL_UNKNOWN" to any casual rides without a User ID to keep the data clean without mixing up subscriber and casual records.

Key Finding: No missing User IDs were found so no changes were needed here.

Q3 — Negative Trip Durations

Negative durations usually happen because of timestamp logging errors. I checked for any trips where the end time was before the start time.

Key Finding: No negative durations found. All trip durations are valid.

Q4 — Station Name Standardization

Station names sometimes have inconsistent spellings like "5th Ave" vs "Fifth Avenue" which can cause problems when grouping by station.

Key Finding: No naming issues found. All 200 station names are consistent.

Overall the dataset was already very clean. All four checks came back with no issues which means the data collection system is working well. This gave me confidence to move straight into the analysis.

5. Stage 3 — Demand and Usage Patterns

This stage looks at when, where, and how bikes are being used. Understanding demand patterns helps the business decide where to put more bikes, when to restock stations, and how to manage the fleet efficiently.

Q1 — Which hours of the day are busiest?

I grouped all trips by the hour they started to find out which hours had the most rides. Hours above the daily average were labeled Peak.

Hour	Total Trips	Label	Hour	Total Trips	Label
0:00	2,030	Off-Peak	12:00	2,086	Peak
8:00 AM	2,145	Peak	13:00	2,061	Off-Peak
9:00 AM	2,084	Peak	17:00	2,104	Peak
10:00 AM	2,117	Peak	18:00	2,030	Off-Peak
11:00 AM	2,107	Peak	20:00 PM	2,126	Peak

Key Finding: 8 AM is the single busiest hour with 2,145 trips — morning commute. 8 PM is the second peak with 2,126 trips — evening leisure rides. Demand is spread fairly evenly across all 24 hours.

Q2 — How do weekday and weekend rides differ?

I compared total trips and average trip duration between weekdays and weekends to understand whether people ride more for commuting or leisure.

Day Type	Total Rides	Avg Trip Duration	Purpose
Weekday	35,494 (71%)	31.93 mins	Work Commute
Weekend	14,506 (29%)	32.12 mins	Leisure

Key Finding: Weekdays have 71% of all rides which makes sense for a commuter-heavy platform. Weekend trips are slightly longer by about 0.2 minutes — people are less rushed.

Q3 — Which stations have too many or too few bikes?

I calculated net bike flow per station by subtracting departures from arrivals. A positive number means bikes are piling up. A negative number means bikes are running out.

Station	Departures	Arrivals	Net Flow	Status
Station 1	211	276	+65	Surplus
Station 175	252	316	+64	Surplus
Station 105	216	278	+62	Surplus
Station 148	275	212	-63	Deficit
Station 102	278	216	-62	Deficit
Station 173	279	225	-54	Deficit

Key Finding: Station 1 has the biggest surplus (+65 bikes) and Station 148 has the biggest deficit (-63 bikes). Moving bikes from Station 1 to Station 148 would fix both problems at once.

Q4 — Are peak hour trips different from off-peak trips?

Time Period	Avg Duration	Avg Distance	Ride Type
Peak Hours	32.24 mins	5.28 km	Commute
Off-Peak Hours	31.90 mins	5.26 km	Leisure

Key Finding: Peak and off-peak trips are almost identical in length and distance. The difference is less than 0.5 minutes which means the same bikes work well at all times of the day.

Q5 — Which stations peak at unusual hours?

Most stations peak during morning (7-9AM) and evening (5-7PM) commute hours. I identified stations that peak outside these windows since they need a different stocking schedule.

Station	Peak Hour	Rides at Peak	Possible Reason
Station 112	2:00 PM	24	Near university or shopping area
Station 114	10:00 AM	23	Midday leisure zone
Station 148	3:00 AM	22	Near nightlife area
Station 188	1:00 AM	22	Late night entertainment zone
Station 197	3:00 AM	22	Night shift area

Key Finding: 13 stations have unusual peak hours. Stations 148, 188 and 197 peak at 1-3 AM suggesting they are near nightlife areas. These stations cannot follow the standard restocking schedule.

6. Stage 4 — User Behavior and Segmentation

This stage focuses on understanding the different types of riders. There are three ride types — Casual (pay per ride), Monthly subscribers, and Yearly subscribers. Knowing how they ride differently helps with marketing and pricing decisions.

Q1 — How do the three rider types compare?

Ride Type	Total Trips	Share	Avg Duration	Avg Distance
Casual	25,074	50.15%	32.00 mins	5.27 km
Monthly	14,977	29.95%	31.99 mins	5.25 km
Yearly	9,949	19.90%	31.96 mins	5.27 km

Key Finding: Casual riders make up exactly half of all trips. All three rider types have nearly identical trip durations and distances — the only difference is how they pay.

Q2 — Who are the most active riders?

I ranked users by total trips and total distance to find the most engaged riders on the platform.

User ID	Ride Type	Total Trips	Total Distance
4647	Casual	14	81.7 km
3611	Casual	14	55.1 km
4378	Casual	13	80.5 km
853	Casual	13	69.7 km
4483	Casual	13	68.9 km
3168	Casual	13	63.0 km

Key Finding: Every single one of the top 10 most active riders is a Casual user. They are riding frequently but paying per-ride rates which are more expensive than a subscription. These users are the best candidates for a subscription conversion campaign.

Q3 — When do different rider types prefer to ride?

I looked at ride counts by hour for each rider type to check if there were any time preferences between casual and subscriber riders.

Hour	Casual	Monthly	Yearly	Total
0:00 AM	1,089	566	375	2,030

Hour	Casual	Monthly	Yearly	Total
8:00 AM	~1,070	~555	~365	~1,990
12:00 PM	~1,045	~620	~415	~2,080
8:00 PM	~1,060	~640	~420	~2,120

Key Finding: Casual riders are the biggest group at every single hour of the day. Usage is 24/7 across all rider types. Yearly subscribers have the lowest counts but they are consistent throughout the day.

7. Stage 5 — Maintenance Analysis

This stage looks at bike maintenance — what types of issues come up most often, how much downtime they cause, and which bikes and stations are most affected. The goal is to help the business plan maintenance better and reduce unexpected breakdowns.

Q1 — What are the most common issues and when do they happen?

Issue Type	Total Cases	Most Affected Station	Peak Season
Flat Tire	2,586	Station 16	Spring
Brake Issue	2,527	Station 182	Summer
Chain Issue	2,447	Station 16	Winter
Battery Issue	2,440	Station 106	Spring

Key Finding: Flat Tire is the most common issue with 2,586 cases peaking in Spring. Station 16 appears twice in this list suggesting the terrain around it causes extra wear. Battery issues peak in Spring — probably because batteries drain over winter and are not checked before the busy season starts.

Q2 — How much does maintenance affect bike availability?

I calculated what percentage of each station's bike fleet was under maintenance to find out which stations were at risk of running out of available bikes.

Station	Bikes in Maintenance	Fleet Affected	Status
Station 171	12	70.59%	Critical
Station 70	9	64.29%	Critical
Station 2	8	50.00%	Critical
Station 12	9	37.50%	High
Station 138	9	36.00%	High
Station 189	11	23.40%	Moderate
Station 49	9	20.00%	Moderate

Key Finding: Station 171 has 70.59% of its bikes in maintenance — that means only about 3-4 bikes are actually available for riders. Three stations are above 50% which is a serious problem. Bikes from surplus stations should be moved here urgently.

Q3 — Which bikes have the most maintenance history?

I looked at which bikes had the highest number of maintenance records and compared that with their average trip distance and duration.

Bike ID	Avg Distance	Avg Duration	Maintenance Count
504	5.49 km	30.65 mins	20
971	5.49 km	29.82 mins	20
394	5.06 km	32.86 mins	20
726	4.92 km	35.00 mins	20
760	4.78 km	29.77 mins	20

Key Finding: Five bikes have hit 20 maintenance records each. Bike 726 is especially concerning — it takes 35 minutes to cover just 4.92 km which suggests ongoing mechanical issues. These bikes should be inspected to decide if they are worth repairing or should be replaced.

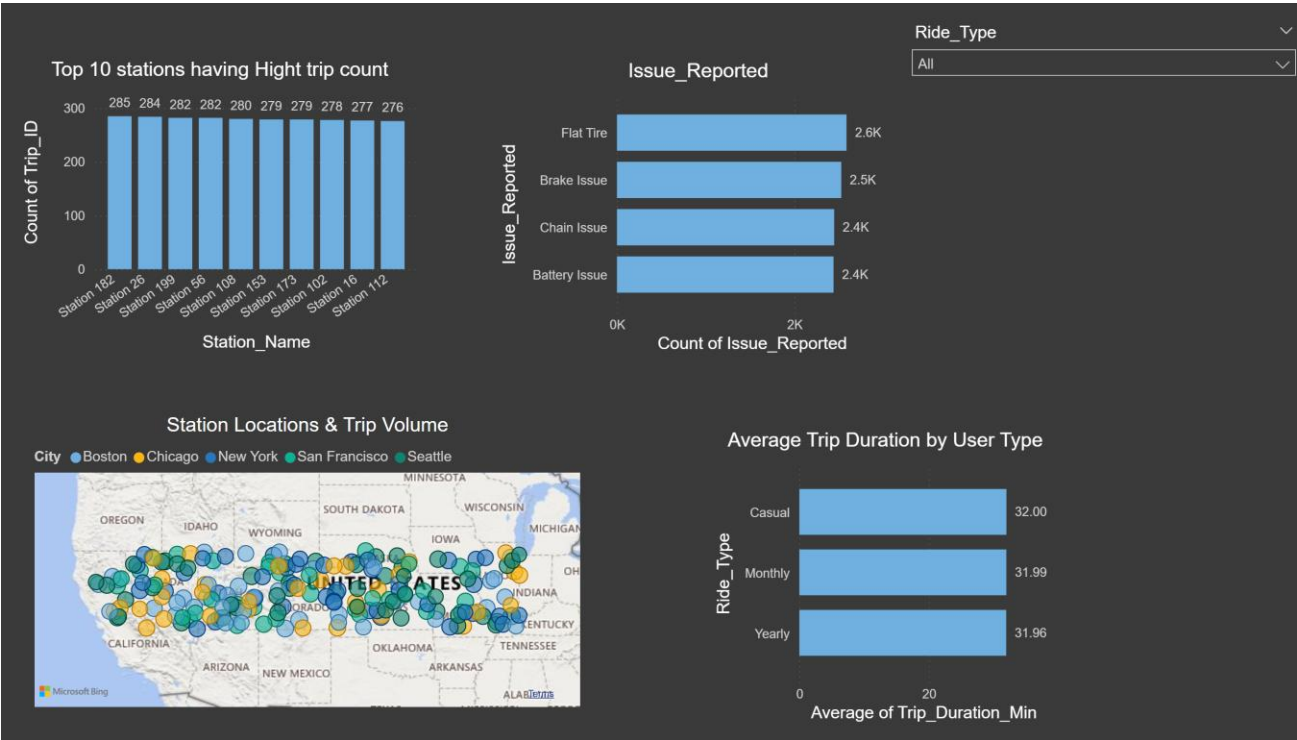
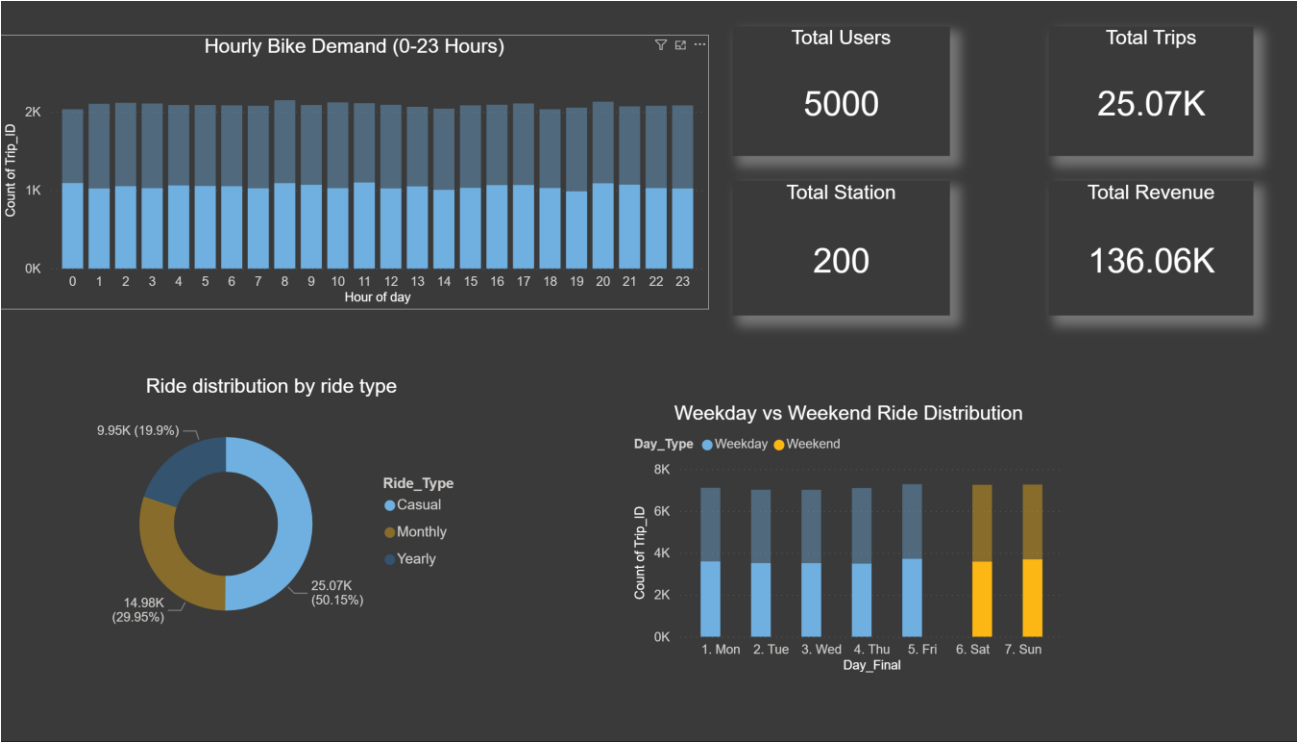
8. Power BI Dashboard

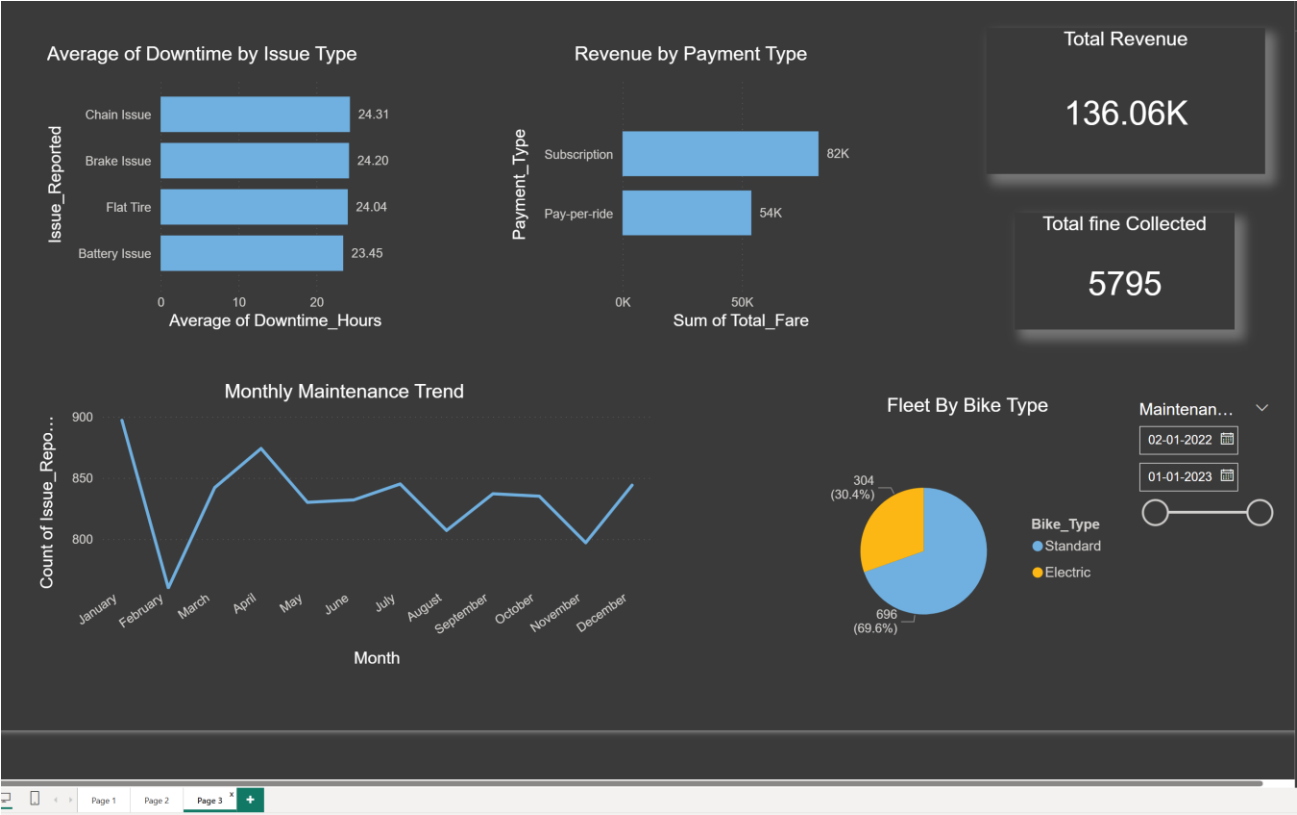
After completing all the SQL analysis I built a 3-page Power BI dashboard to present the findings visually. The dashboard connects directly to the MySQL database and uses slicers so users can filter by ride type or city interactively.

Page	Title	Visuals
1	Overview	KPI Cards (trips, users, stations, revenue), Hourly Demand Chart, Rides by User Type Donut, Weekly Trip Pattern
2	Station and Demand	Top 10 Busiest Stations, Maintenance Issues Bar Chart, Station Map across 5 cities, Avg Duration by Ride Type, Ride Type Slicer
3	Maintenance and Revenue	Avg Downtime by Issue, Revenue by Payment Type, Monthly Maintenance Trend, Total Revenue and Fines Cards, Fleet by Bike Type

Key Numbers from the Dashboard

Metric	Value	Metric	Value
Total Trips	50,000	Total Revenue	136,060
Total Users	5,000	Total Fines	5,795
Casual Rider Share	50.15%	Subscription Revenue	82K (60%)
Peak Hour	8:00 AM	Pay-per-ride Revenue	54K (40%)
Busiest Station	Stn 182	Electric Bikes	30.4%
Top Issue	Flat Tire	Standard Bikes	69.6%





9. Recommendations

Based on everything found in this analysis, here are some practical steps the business could take to improve operations and grow revenue.

1. Rebalance Bikes Between Stations

Station 1 has 65 more bikes arriving than leaving while Station 148 has 63 fewer. Moving bikes from surplus stations to deficit ones would directly fix shortages. Setting up an alert when net flow goes beyond +/- 30 would help catch imbalances before they become a real problem for riders.

2. Target Active Casual Riders with Subscription Offers

All top 10 most active riders are casual users paying per ride. A simple message like "You have taken 14 rides this month, save money with a monthly plan" could convert them into subscribers. This is probably the easiest revenue improvement available.

3. Create Different Stocking Schedules for Unusual Stations

13 stations peak outside normal commute hours. Stations near nightlife areas need bikes restocked overnight. University area stations need bikes in the mid-morning. Using the same restocking schedule for all 200 stations is not working well for these locations.

4. Plan Maintenance Around Seasons

Flat tires peak in Spring and brake issues peak in Summer. The business should stock up on spare tires before Spring and do brake checks before Summer every year. Battery checks should happen in late Winter before the riding season picks up.

5. Urgent Action Needed at Station 171

Over 70% of bikes at Station 171 are in maintenance right now. That means barely any bikes are available for riders there. This needs immediate attention — either move bikes from nearby surplus stations or prioritize repairs at this location first.

6. Consider Retiring High-Maintenance Bikes

Five bikes have each been repaired 20 times. At some point it makes more financial sense to replace a bike than keep fixing it. A policy to review bikes after 15 repairs would prevent money being spent on bikes that keep breaking down.

7. Grow the Electric Bike Fleet

Only 30% of bikes are electric right now. Given that sustainability is increasingly important in urban transport, increasing this to around 50% could improve both the rider experience and long-term revenue.

10. Summary and Conclusion

This project covered a complete end-to-end analysis of a bike-sharing platform using MySQL and Power BI. Starting from data quality checks all the way to maintenance patterns and revenue breakdown, the analysis covered 18 business questions across five stages.

A few things stood out the most. First, the dataset was surprisingly clean with no missing values or duplicates. Second, casual riders account for half of all trips but also make up all of the top 10 most active users, which is a big revenue opportunity. Third, some stations are in serious trouble with over 70% of their bikes in maintenance.

Stage-by-Stage Summary

Stage	Questions	Main Finding
Stage 1 — Quality Check	3	Dataset is 100% complete — no missing values or gaps
Stage 2 — Cleaning	4	No duplicates or anomalies found
Stage 3 — Demand	5	Peak at 8AM and 8PM, 13 stations with unusual patterns
Stage 4 — Users	3	Casual riders = 50% of trips, top riders are all casual
Stage 5 — Maintenance	3	Station 171 at 70% downtime, 5 bikes need replacement review
Total	18	7 actionable recommendations identified

*Analysis done using MySQL for data extraction and Power BI for dashboard.
Dataset covers 50,000 trips across 7 tables and 5 US cities.*